Keshav Aggarwal

Personal Details

Email	keshavagg1098@gmail.com
Phone	+91 94586 66775
Address	Department of Astronomy, Astrophysics and Space Engineering, IIT Indore, Simrol, Indore 453552, India
Find me	Webpage, Google Scholar, ORCID, NASA-ADS, ResearchGate, LinkedIn

Education

Jul 2023 – May 2026	Ph.D. Astronomy – **DAASE, Indian Institute of Technology (IIT) Indore** (thesis submitted May 2026) Funding agency: Prime Minister's Research Fellowship Collaborating agency: Space Physics Laboratory, Vikram Sarabhai Space Centre, Thiruvananthapuram Supervisor: Prof. Abhirup Datta *Exploring space weather and its effect on planetary atmospheres using radio occultation*
Aug 2021 – Apr 2023	M.Sc. Astronomy – **DAASE, Indian Institute of Technology (IIT) Indore** Supervisor: Prof. Abhirup Datta *Study of the solar corona using the Indian Mars Orbiter Mission*

Research Positions

Oct 2023 – Present	**Ph.D. Student Visitor**, Space Physics Laboratory, Vikram Sarabhai Space Centre (ISRO), Thiruvananthapuram
Jul 2023 – May 2026	**Prime Minister's Research Fellow**, DAASE, IIT Indore

Workshops, Schools, and Presentations

Mar 2026	"*A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening*" at **7th Indian Planetary Science Conference (IPSC 2026), IIT Indore, India** (Talk)
Feb 2026	"*Estimating solar wind velocities using spacecraft radio signals*" at **National Space Science Symposium (NSSS 2026), Umiam, Shillong, India** (Talk)
Jan 2026	"*Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki*" at **ISRO–ESA Workshop, Thiruvananthapuram, India** (Poster)
Dec 2025	"*Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki*" at **AGU Fall Meeting 2025, New Orleans, United States** (Poster)

Dec 2025 — *"Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki"* at **The Variable Sun, Thiruvananthapuram, India** (Poster)

Sep 2025 — *"Probing the Smiley-Sun using Akatsuki radio signals"* at **ASI Symposium 003, Astronomical Society of India, JECRC University, Jaipur, India** (Poster)

Aug 2025 — *"Studying the 'Smiley Sun' using the Akatsuki radio occultation experiment"* at **URSI Asia-Pacific Radio Science Conference (URSI AP-RASC 2025), International Convention Centre, Sydney, Australia** (Talk)

Mar 2025 — *"On the estimation of solar wind velocity using spectral characteristics of the probing radio signals: results from solar occultation measurements using Indian Mars Orbiter Mission"* at **6th Indian Planetary Science Conference (IPSC), IIT Roorkee, India** (Poster)

Oct 2024 — *"Results from radio occultation studies using Indian Mars Orbiter Mission"* at **6th URSI Regional Conference on Radio Science (URSI-RCRS), Graphic Era Hill University, Bhimtal, India** (Poster)

Oct 2024 — *"Results from radio occultation studies using Indian Mars Orbiter Mission"* at **3rd Indian Space Weather Conference, IIT Roorkee, India** (Talk)

Oct 2023 — *"Unveiling space weather and planetary atmosphere dynamics through Aditya-L1 and DISHA data integration"* at **2nd Indian Space Weather Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)

Sep 2023 — *"Retrieving sulphuric acid profiles of the Venus atmosphere from Akatsuki radio occultation data"* at **Venus Science Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)

Aug 2023 — **Strange New Worlds conference**, Indian Institute of Science Education and Research (IISER), Pune

Feb 2023 — **ISRO–ARIES Aditya-L1 Support Cell Workshop 3**, Indian Institute of Technology (BHU), Varanasi

Jul 2022 — **IIA Summer School 2022**, Indian Institute of Astrophysics (IIA), Bengaluru

Jun 2022 — **Introductory Summer School in Astronomy and Astrophysics 2022**, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune

Mar 2022 — **Spring 2022: ERS Post-Launch Data Challenge for JWST**, Space Telescope Science Institute, Baltimore, and Heidelberg

Sep 2021 — *"HR diagrams of star clusters"* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)

Aug 2021 — *"The Gaia telescope and the use of its data"* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)

Feb 2021 — *"Life in the Solar System, and exoplanets"* at **Nakshatra, Astronomy Club, IIT Indore, Indore, India** (Talk)

Achievements

Dec 2025	Recipient of the *AGU Student Travel Grant* for presenting a poster at AGU Fall Meeting 2025, New Orleans, USA
Mar 2025	*Early Career Researcher Award for best poster* at the 6th Indian Planetary Science Conference, IIT Roorkee
Oct 2023	*Prime Minister's Research Fellowship*, Ministry of Education, Government of India

Internships

Aug 2022 – Sep 2023	**Researcher in AI and Orbital Mechanics | HR Coordinator** at ERETS Space London, London (remote) Developed AI/ML and orbital mechanics algorithms for satellites. Handled hiring for the team.
Feb 2022 – Aug 2022	**Data Analyst | Project Coordinator** at Armstrong Space Australia, Melbourne (remote) Built pipelines for acquiring and processing space debris data. Space debris removal by laser ablation. Catalogues of debris between 1 and 10 cm.

First-Author Publications

1. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Manikantan, R., Sharma, A., Imamura, T., *A study on the contribution of the interplanetary medium in radio occultation experiments*, **Icarus**, 2026, 10.1016/j.icarus.2026.117291.
2. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Sharma, A., *Plasma turbulence in the lunar environment across solar wind and magnetotail conditions: Observations from Chandrayaan-2 Radio Science experiment*, **Journal of Geophysical Research: Planets**, 2026, 10.1029/2026JE009957.
3. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Imamura, T., Ando, H., *A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening*, **Monthly Notices of the Royal Astronomical Society**, 2026, 10.1093/mnras/stag554.
4. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Roopa, M. V., Imamura, T., *A generalized method for estimating solar wind speeds and densities using spectral broadening for a Kolmogorov turbulence spectrum*, **Advances in Space Research**, 2026, 10.1016/j.asr.2026.03.059.
5. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Imamura, T., *On the estimation of solar wind velocity under varying solar activity conditions using Akatsuki measurements*, **Monthly Notices of the Royal Astronomical Society**, 2025, 10.1093/mnras/staf1305.

6. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Roopa, M. V., Dai, B. K., *Insights into solar wind flow speeds from the coronal radio occultation experiment: Findings from the Indian Mars Orbiter Mission*, **The Astrophysical Journal**, 2025, 10.3847/1538-4357/adb627.
7. **Aggarwal, K.**, Datta, A., Choudhary, R. K., *Studying the 'Smiley Sun' using the Akatsuki radio occultation experiment*, **Proceedings of the 7th URSI Asia-Pacific Radio Science Conference (AP-RASC 2025)**, 2025, 10.46620/ursiaprasc25/ahct9245.
8. **Aggarwal, K.**, Choudhary, R. K., Datta, A., *Results from radio occultation studies using Indian Mars Orbiter Mission*, **Proceedings of the 6th URSI Regional Conference on Radio Science (RCRS 2024)**, 2024, 10.46620/ursi_rsrc24/1017qon9346.